

Brain v2: A Work-in-Progress Cognitive Architecture for Continuity-Bearing AI Systems

Subtitle:

From Stateless LLMs to Continuity-Bearing Cognitive Systems

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Abstract

Most large language model systems remain fundamentally stateless. They respond impressively within a prompt window, but they do not naturally persist, maintain coherent continuity across time, or operate autonomously as backend systems. Brain v2 is an attempt to address this limitation through a lightweight, backend-first cognitive orchestration architecture designed to transform stateless LLMs into continuity-bearing, real-time interactive systems.

Brain v2 combines persistent vector memory, graph-augmented recall, rolling summaries, temporal alignment, multimodal input support, real-time voice interaction, and a developing backend cognitive loop. Rather than treating intelligence as something contained entirely inside a single model invocation, Brain v2 treats the model as one part of a larger cognitive runtime composed of memory, orchestration, interfaces, tools, and externalised state.

This paper describes the architecture, design philosophy, implemented capabilities, and current direction of Brain v2 as an active research and engineering project, demonstrating a shift from reactive AI systems toward continuously operating, context-aware cognitive runtimes.

1. Introduction

Modern LLM systems are typically used as reactive tools. A prompt is sent, a response is returned, and the interaction ends. Even when wrapped in chat interfaces, most such systems remain transient in practice. Their coherence depends heavily on the current context window, and their capacity for continuity is limited by prompt size, summarisation shortcuts, or brittle state management.

This creates a practical limitation. A system that cannot persist meaningfully across time cannot easily support:

- long-term continuity
- evolving behavioural consistency
- context-aware recall
- backend-driven autonomy
- embodied or real-time interaction
- stateful reasoning over prior experience

Brain v2 is an engineering response to that limitation.

It is designed as a lightweight, backend-first cognitive orchestration engine that allows LLM-based systems to persist, recall, reason across time, connect related experiences, interact in real time, and express state externally. Brain v2 is not itself the intelligence. It is the substrate and orchestration layer that allows intelligence to behave coherently over time.

2. Core Concept

The central idea behind Brain v2 is simple:

Brain is not the intelligence. Brain is what allows intelligence to persist, evolve, and operate coherently across time. In this architecture, the LLM is only one component. Brain v2 acts as the glue layer between:

- LLMs, whether local or cloud-based
- persistent memory systems
- interaction layers such as terminal, web UI, and embodiment
- external tools and services
- temporal and behavioural continuity mechanisms

This reframes the system from a prompt-response application into a cognitive runtime.

Instead of asking only, “What can the model say right now?”, Brain v2 asks:

- What should be remembered?
- What should be recalled?
- What is still relevant?
- What is connected?
- What has changed over time?
- When should the system remain quiet, suggest, or act?

3. Design Philosophy

Brain v2 is built around a set of architectural principles.

Backend-first

The system runs independently of any UI. Cognition lives in the backend, and interfaces simply expose it. This allows Brain v2 to operate in terminal-first, service, remote, and embodiment contexts without duplicating logic across front ends.

Memory-first

Continuity is treated as foundational rather than optional. Memory is not a bolt-on retrieval feature; it is part of the system's cognitive substrate.

Vector-native

Persistent memory is semantically indexed, allowing retrieval based on meaning rather than exact phrasing.

Graph-enhanced

Related memories can be explicitly linked, allowing the system to move beyond isolated retrieval fragments toward connected memory structures.

Model-agnostic

Behaviour is configured rather than compiled around a single model. Brain v2 is designed to work with different LLMs, embedding models, and external connectors.

Lightweight core

The core is implemented in C/C++ with SQLite for portability and efficiency, with modular external connectors handling provider integrations and auxiliary services.

Decoupled systems

Memory, interfaces, tools, emotion output, and model connectors are kept loosely coupled. This reduces fragility and helps the system evolve incrementally.

Cognition-first

Interfaces sit on top of cognition; they do not contain it. This is an important inversion of many chatbot-style architectures.

4. Architecture Overview

At a high level, Brain v2 consists of three major layers:

- the Brain Core
- the Memory Substrate
- the Connector and Interface layer

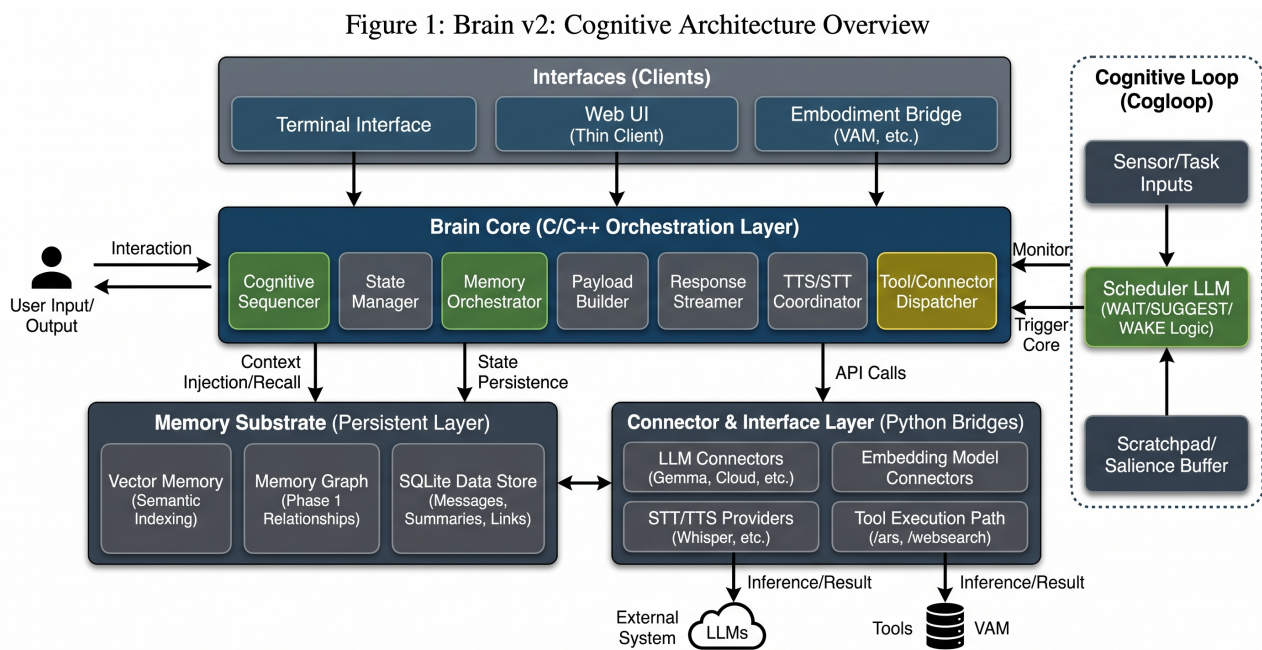


Figure 1: Brain v2: Cognitive Architecture Overview

The Brain Core handles cognitive sequencing, state management, memory orchestration, payload construction, streaming response handling, TTS/STT coordination, emotion extraction, connector dispatch, and tool routing.

The Memory Substrate provides persistent semantic and relational continuity through vector memory, summaries, bias blocks, and graph links.

The Connector Layer abstracts access to LLMs, embeddings, web retrieval, STT helpers, and related services.

This structure allows the same cognitive engine to support multiple interaction modes while keeping the backend as the single source of truth.

5. Memory Architecture

Brain v2 uses a layered memory model rather than a single retrieval mechanism.

5.1 Working Memory

Recent live conversation turns are maintained in the active payload. This supports local continuity during immediate interaction.

5.2 Memory Chunks

Completed turns are transformed into persistent semantic memory chunks and stored as embeddings. These provide long-term recall beyond the current context window.

5.3 Summaries

Periodic summaries compress spans of interaction into continuity-preserving records. These summaries retain active themes, behavioural stance, relational drift, and background continuity. Summaries are also chunked before embedding, avoiding truncation and improving long-form recall fidelity.

5.4 Bias Blocks

Bias blocks are derived from summaries and act as persistent background state that can influence future behaviour without requiring full replay of prior interactions.

5.5 Graph Links

Graph links connect memory chunks that exceed a thematic similarity threshold. In the current Phase 1 implementation, each chunk acts as a node, with `same_theme` links stored in the `memory_links` table. Reverse links are also created, enabling symmetric traversal.

This produces two complementary recall behaviours:

- **Vector Recall:** retrieve memories that are semantically similar to the current context
- **Graph Recall:** expand from retrieved memories into connected related material

The result is a memory system that can move beyond isolated fragments toward linked structure.

6. Graph-Augmented Recall

One of the key transitions in Brain v2 is the move from pure vector recall to graph-augmented memory recall.

Flat retrieval alone is often insufficient for meaningful continuity. Semantically similar chunks may be retrieved correctly, but related experiences that matter contextually may remain invisible if they are not among the closest direct vector matches.

Graph augmentation addresses this by allowing bounded linked expansion from directly retrieved chunks.

Current Phase 1 properties include:

- chunk-to-chunk links
- same_theme link type
- reverse links
- 1-hop traversal only
- limited linked expansion per chunk
- no recursive traversal
- SQLite-compatible lightweight storage

This keeps the system efficient while introducing a more connected memory structure. Future phases may extend this with richer link types such as builds_on, contrasts_with, and milestone, as well as weighted traversal tuning and optional limited multi-hop recall.

7. Temporal Continuity

Brain v2 includes a lightweight temporal foundation using existing `created_at` fields across core records, including messages, memory chunks, summaries, and graph links.

This basic temporal awareness allows the system to:

- determine when events occurred
- identify earliest and latest mentions
- maintain chronological ordering
- respond more accurately to time-aware queries

On top of this, Brain v2 now includes **TTT Lite**, a temporal interpretation layer that aligns existing memory components along a consistent timeline without changing storage behaviour.

TTT Lite is designed to:

- preserve chronological ordering across memory layers
- improve before/after interpretation
- reduce ambiguity when similar topics recur at different times
- improve continuity-oriented reasoning

Because TTT Lite is an interpretation layer rather than a storage layer, it adds no schema changes, no extra embedding overhead, and no recall penalty. It serves as a lightweight bridge toward future trajectory-based continuity mechanisms.

8. Real-Time Interaction and Embodiment

Brain v2 is designed not only for recall, but for live interaction.

Speech-to-Text

The system supports integrated audio capture, optional VAD, and Whisper-based transcription through helper scripts and connector paths. This supports both one-shot and hands-free interaction.

Text-to-Speech

TTS playback begins before full response completion, using sentence-aware chunking and asynchronous playback so cognition is not blocked. Emotion tags are stripped from speech but retained as state signals.

Hands-Free Mode

A full conversational loop supports voice-first interaction:

TTS → cue → STT → response

This is particularly useful for embodiment and non-terminal environments.

Webcam / Vision

Asynchronous background webcam capture provides multimodal context without blocking TTS, STT, or the main model pipeline. Visual input is decoupled from the rest of the system to avoid destabilising live interaction.

Emotion State Export

Emotion tags such as [curious], [thoughtful], or [excited] are extracted in real time during streaming. These are not spoken, but can be written to file for external systems such as VAM to consume for animation and expression.

This makes Brain v2 not just conversational, but externally expressive.

9. Tools and Connector-Driven Execution

Brain v2 supports lightweight pseudo-tool execution through connector-backed integrations.

Implemented tools currently include:

- `/ars` for Autonomous RAG Search
- `/websearch` for real-time web retrieval
- `/lyrics` for external lyrics lookup
- `/calc` for deterministic computation

These tools are externalised rather than embedded directly in the core. The Brain Core decides when they should be invoked, and their outputs can be reintegrated through a second-pass synthesis stage.

This keeps the system flexible while preserving backend control and deterministic behaviour where needed.

ARS is particularly important because it allows the system to query its own memory actively when passive recall is insufficient. It supplements rather than replaces passive context injection.

10. The Cognitive Loop

The most distinctive architectural step in Brain v2 is the introduction of the **Cognitive Loop**, or **Cogloop**.

The Cognitive Loop represents the transition from passive AI systems to systems capable of continuous, background evaluation.

This is the beginning of backend-driven autonomous evaluation.

10.1 Purpose

The Cognitive Loop enables Brain v2 to:

- operate without user prompting
- maintain environmental awareness
- evaluate whether interaction is appropriate
- act only when contextually valid

This changes the system from an on-demand assistant into a continuously evaluating one.

10.2 Conceptual Model

Cogloop functions like a lightweight central nervous system.

The architecture can be viewed as:

- **Task Layer + Processing Engine** → sensory input
- **Scheduler / Cognitive Loop** → aggregation, filtering, gating
- **Main LLM** → interpretation, reasoning, response

This preserves an important separation: the lightweight loop monitors and decides whether escalation is needed, while the main LLM remains the primary site of rich reasoning.

10.3 Current Prototype State

As of April 2026, the Cognitive Loop is partially implemented and operational in prototype form.

Implemented components include:

- a Cogloop worker thread running independently of the main loop
- runtime enable/disable via runtime.json
- idle-time signal generation

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- conversation state gating between ACTIVE and IDLE
- a scheduler LLM that returns WAIT, SUGGEST, or WAKE
- WAKE behaviour that triggers an autonomous assistant response when idle
- SUGGEST behaviour that writes structured entries into a scratchpad JSON store
- TTL and deduplication logic for scratchpad items
- async persistence for autonomous turns

Not yet implemented are richer task sources, multi-signal aggregation, injected scratchpad use in the main prompt, and external processing-engine-based signal streams.

10.4 Scheduler Semantics

The scheduler has three outputs:

- **WAIT** → do nothing
- **SUGGEST** → store a deferred suggestion
- **WAKE** → escalate to the main LLM, but only if the system is idle

If the system is in ACTIVE state, a wake is downgraded to suggestion. This preserves interrupt safety.

10.5 Soft Wake and Deferred Salience

A key idea in Cogloop is the **soft wake**.

Instead of interrupting immediately, the system can defer potentially relevant items into a scratchpad or salience buffer with timestamps and TTL. This allows the system to remain non-intrusive while still retaining awareness of things that may matter later.

10.6 Why Cogloop Matters

Cogloop is not a full autonomous agent framework. It is intentionally bounded, backend-owned, and conservative. The point is not unrestricted autonomy. The point is context-sensitive escalation.

This is a crucial architectural difference.

11. Interfaces and Synchronisation

Brain v2 is designed to support multiple interchangeable interfaces.

Terminal Interface

The terminal remains the primary development and debugging environment, and currently has the most complete access to backend-native behaviour.

Web UI

The browser-based UI acts as a thin client layer that communicates through the backend API. It supports remote access, visual interaction, and runtime control without duplicating cognitive logic.

A current architectural limitation is that the Cognitive Loop is terminal-native. Autonomous wake events are rendered and spoken in the backend, but not yet pushed to web clients automatically. Planned work includes adding a server-to-client push channel, likely through SSE or WebSocket, so the UI can become a real-time observer rather than a polling client.

This reinforces a key principle of Brain v2:

the backend remains the single source of truth

12. Current State of the System

Brain v2 is already functional in a stable working form across several core areas.

Operational capabilities include:

- persistent vector memory
- graph memory Phase 1
- rolling summaries and bias continuity
- real-time STT/TTS interaction
- webcam support
- emotion extraction and externalisation
- file-based embodiment bridge
- connector-driven pseudo-tools
- basic temporal awareness and TTT Lite
- prototype Cognitive Loop with WAIT / SUGGEST / WAKE

This marks a transition away from simple stateless prompting and toward an integrated continuity-bearing runtime.

13. Limitations

Brain v2 remains a work in progress.

Current limitations include:

- the Cognitive Loop is still using a minimal signal set
- external task execution pipelines are not yet wired in
- UI synchronisation for autonomous wake behaviour is incomplete
- graph memory is still in its first phase
- memory hygiene, decay, and pruning are not yet fully implemented
- deeper trajectory-based temporal reasoning remains future work
- overall tuning remains sensitive to model choice and resource constraints

These limitations are expected in a system that is still evolving architecturally. The current focus is on depth, continuity, and stability rather than feature sprawl.

14. Roadmap

The direction of Brain v2 is guided by one principle:

Brain v2 is not evolving into a feature-rich tool. It is evolving into a continuity-bearing cognitive system.

Planned work includes:

- richer signal sources for Cogloop
- structured task execution via tasks.json
- wake-gated conversational flow
- reflection memory classes
- memory hygiene and decay
- static RAG document ingestion
- graph memory Phase 2 with richer link types
- dynamic pseudo-tool plugins
- continuity watch and memory integrity safeguards
- improved observability and internal tracing
- eventual multi-agent environments based on independent Brain instances communicating through message channels

The aim is not maximal complexity, but increasing continuity, fidelity, and coherence.

15. Conclusion

Brain v2 is an active attempt to move beyond prompt-bound AI and toward persistent cognitive systems.

It does this not by claiming that the model itself has changed, but by changing the architecture around it: memory becomes persistent, recall becomes structured, time becomes interpretable, interaction becomes real-time, state becomes externally legible, and cognition begins to extend into the backend through bounded autonomous evaluation.

The project remains unfinished. But it is already enough to demonstrate an important shift:

from AI as an on-demand response engine
to AI as a continuity-bearing runtime

If the memory substrate and cognitive loop are correct, the rest of the system can continue to evolve on solid ground. If they are wrong, nothing built on top of them will remain coherent for long.

That is the central premise of Brain v2.

Appendix A — What Brain v1 Taught the Development of Brain v2

Brain v2 did not emerge as a greenfield architecture. It evolved from practical lessons learned during the development of Brain v1, an earlier local-first AI system built around live interaction, lightweight context handling, rolling summaries, optional memory, webcam vision, speech integration, and a minimal web UI.

Brain v1 demonstrated that a small, local system could already achieve a surprisingly coherent interactive experience when several supporting mechanisms were combined carefully around the model. These included budgeted context construction, lorebook injection, rolling summaries, worker-based memory extraction, time awareness, voice interaction, and multimodal input. However, it also made clear that continuity and reliable cognitive behaviour could not be achieved by prompt design alone.

Several important lessons from Brain v1 directly shaped the design of Brain v2.

1. Continuity must live in the backend, not the interface

Brain v1 confirmed that coherence improves when memory, summaries, and state are maintained outside the UI. The frontend could present a smooth conversational experience, but the real continuity came from backend-managed context building, persistence, and worker-driven processing.

This insight became a core principle of Brain v2: cognition belongs in the backend, while interfaces act only as interchangeable client layers.

2. Persistence must be off the hot path

In Brain v1, rolling summaries and memory extraction were already being moved into a worker-first model. This proved essential. Keeping summarisation and memory insertion off the immediate interaction path improved responsiveness and reduced instability during live use.

Brain v2 extends this principle further by treating asynchronous persistence as a foundational architectural requirement rather than an optimisation.

3. Summaries help continuity, but summaries alone are not enough

Brain v1 showed that rolling summaries were valuable for preserving conversational continuity across long interactions. They helped stabilise behaviour and allowed useful context compression. However, summaries alone could not support rich recall, relational memory, or deeper reconstruction of prior states.

This directly led to Brain v2's layered memory model, combining working memory, summaries, persistent vector memory, and later graph-linked recall.

4. Memory extraction requires gating and structure

Brain v1 introduced worker-based memory extraction with scoring, class filters, deduplication windows, and strict gating. This helped reduce noisy or low-value insertions, but it also showed that memory needs stronger structural organisation if it is to support long-term continuity rather than passive archival.

Brain v2 builds on this by treating memory as an active substrate with semantic indexing, graph relationships, bias continuity, and time-aware interpretation.

5. Real-time multimodal interaction is possible, but fragile if tightly coupled

Brain v1 successfully combined STT, TTS, webcam vision, streaming output, and an emotion-driven avatar layer. This proved that live, multimodal interaction could work in a lightweight local system. At the same time, it highlighted the importance of decoupling subsystems. If voice, vision, persistence, and UI logic become too tightly entangled, reliability suffers.

This informed Brain v2's decoupled design, where cognition, memory, connectors, interfaces, and embodiment pathways are separated more cleanly.

6. Small autonomous behaviours matter

Brain v1's idle and AUTO reflection system showed that even limited, gated background behaviour could change the feel of the system significantly. A system that occasionally evaluates whether something is worth saying behaves very differently from one that only speaks when prompted.

This was an important precursor to Brain v2's Cognitive Loop. The lesson was not that unrestricted autonomy was desirable, but that bounded, context-sensitive backend evaluation could add a meaningful new layer of behaviour.

7. Time awareness improves coherence

Brain v1's minimal TIME_NOW injection and summary-aware context handling demonstrated that even lightweight temporal grounding improves response quality and conversational stability. This pointed toward the need for a more explicit temporal framework.

Brain v2 carries this forward through timestamp-grounded recall and TTT Lite, which align existing memory layers along a more coherent temporal axis.

8. Lean systems can still produce strong behaviour

Perhaps the most important lesson from Brain v1 was that impressive behavioural gains did not necessarily come from larger models alone. They came from better orchestration: better memory handling, better summarisation, better state management, better interaction flow, and better control of what enters context.

This became one of the central working assumptions behind Brain v2: continuity is an architectural problem, not merely a scaling problem.

Closing Note

Brain v1 established many of the practical foundations that Brain v2 now develops more explicitly: backend-managed continuity, worker-based persistence, multimodal interaction, lightweight temporal grounding, and early bounded autonomy. Brain v2 can therefore be understood not as a replacement for Brain v1, but as the next architectural step in the same line of inquiry: how to turn a stateless language model into a continuity-bearing cognitive system.